

Precalc Lesson 1.5 Homework

Name _____

1. Identify the parent function. Then describe the transformations that produce the graph of f .

a. $f(x) = |x + 9|$

b. $f(x) = 5x^3 - 8$

2. The graph of $y = x^3$ is shifted right 4, down 2 and then stretched by a factor of 6. Write an equation that represents the resulting graph.

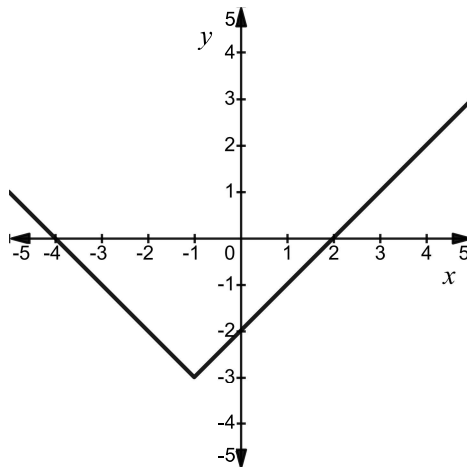
3. Which of the following equations could represent the graph shown?

A) $y = |x - 1| - 2$

B) $y = |x + 1| - 3$

C) $y = -|x + 1| + 2$

D) $y = |-x| + 3$

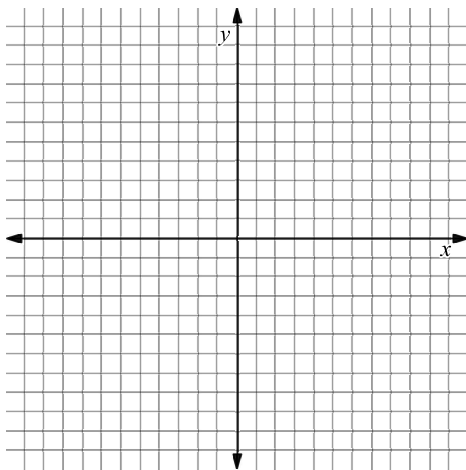


4. Identify the domain and range of $y = -\frac{1}{2}(x - 4)^2 + 6$.

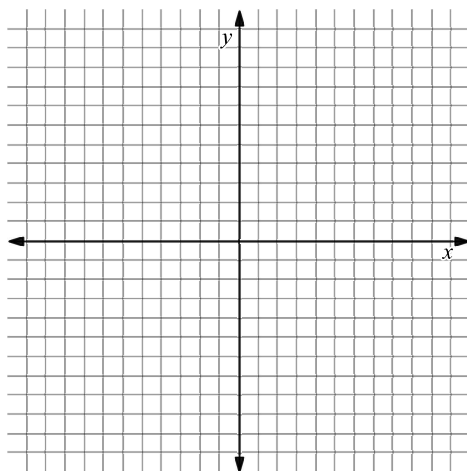
5. Identify the domain and range of $y = 1 + 5\sqrt{x + 4}$.

6. Describe the relationship between the graph of $f(x) = \frac{1}{x-2} + 7$ and the graph of its parent function.

7. Graph $y = 3 + \sqrt{-x}$.



8. Graph $y = \frac{1}{x+5} - 2$.



9. The graph of the function $y = x^3$ was transformed according to the transformations given below, resulting in the graph of the function $y = a(x - b)^3 + c$, for some constants a, b , and c .
- 1) reflection over the x -axis
 - 2) vertical shrink
 - 3) horizontal shift left
 - 4) vertical shift up

Eleanor was asked to provide a possible value for a, b , and c . She said $a = -\frac{1}{3}, b = 4$, and $c = 7$. Determine if each of her values are reasonable and give reasons for your answers.

10. Using the graph of $g(x)$ shown, sketch the transformation $g(-x) + 2$ on the same coordinate grid.

